The smallest grammars for Thue-Morse words

Exact minimum sizes for the Thue-Morse family raised by Mieno, Inenaga and Horiyama [1].

The word and the cost

Start with 0 and repeatedly replace 0 by 01 and 1 by 10. Write T_n for the word after n substitutions, so T_3 = 01101001 and T_n has 2^n bits. A straight-line grammar is an acyclic list of concatenation rules with one start rule. Its cost counts all terminal and rule-reference symbols on the right-hand sides, including the start rule.

Theorem

Let g_n count the fewest right-hand-side symbols. Let c_n count the fewest productions when every rule is a binary concatenation or produces one terminal.

$$g_0 = 1, \qquad g_1 = 2, \qquad g_2 = 4, \qquad g_n = 4n - 5 \quad (n \geq 3).$$

$$c_0 = 1, \qquad c_n = 2n + 1 \quad (n \geq 1).$$

Level n	Bits in the word	Grammar symbols	Binary-model rules
3	8	7	7
4	16	11	9
10	1,024	35	21
20	1,048,576	75	41

Why the lower bound works

After each substitution the original bits sit at the even positions. Every reusable piece of at least four bits reveals whether it starts at an even or an odd position. That lets us delete alternate bits directly from a grammar. A fixed prefix rules out every way to do this while saving fewer than four symbols at both choices of parity.

The proof checks a finite prefix obstruction and the smallest base cases. It then applies the same reduction to every larger word. The encoder constructs a grammar of the matching size.

These counts measure the grammar itself. A serialized file also encodes rule boundaries and references.

Half the bits, four fewer symbols

Normalize a minimum grammar

Delete unreachable rules. Remove empty rules and eliminate aliases with a one-symbol body. Every child now expands to a strictly shorter word than its parent. If two rules expand to the same word, redirect references to one rule and delete the other. Thus expansion words are distinct. Inlining a non-start rule referenced once saves one symbol overall.

Alignment of a reusable word

Index the bits from zero. Equal adjacent bits only occur across a boundary between substituted pairs. In a factor containing an equal pair, that pair therefore fixes the parity of the start. A four-bit alternating factor cannot start at an odd position. Its form would be (1−a), b, (1−b), c for three consecutive source bits a, b, c. Alternation would give a = b = c. Three equal bits cannot occur in a Thue-Morse word because any three consecutive positions contain a complete substituted pair.

Every expansion of length at least four has a fixed start parity. A length-three expansion containing an equal pair also has a fixed start parity.

Project a grammar by dropping parity p

Give each rule of expansion length at least four one projected rule. The fixed phase tells us which bits its children retain. Drop a literal at parity p. Replace a length-two child with its single remaining bit. A length-three child keeps either its middle bit or its two outer bits. The middle bit needs no rule. The outer pair uses a rule of cost two, shared by all uses of that child in this projection. A nonalternating triple starting at p keeps only the middle bit.

Let b count length-two expansions, d count length-three rules with three-symbol bodies, f_p count nonalternating triples starting at p, and t_p count literal slots at parity p in rules with expansions of length at least four. The saving is at least

$$D_p = 2b + d + 2f_p + t_p.$$

The two projections produce T_{n-1} and its bitwise complement, which have equal minimum costs by swapping 0 and 1. Thus $D_p \geq 4$ for either p gives $g_n \geq g_{n-1} + 4$.

The exceptional budgets

Assume both D_0 and D_1 are at most three. The following cases cover every possible set of short expansions and allowed counts of literal slots in longer rules.

One pair rule. Two pair rules would already save four. With one, any triple must alternate and there is at most one literal slot of each parity in longer rules. For pair 01 or 10, we may allow both alternating triples regardless of which are present. For pair 00 or 11, an alternating triple would have a three-literal rule, using up the remaining budget at both parities. With no literal slots in longer rules, the word would be tiled by that pair and alternating triples. None can start its prefix 011. Thus the cases for pairs 00 and 11 allow no triples.

Zero pair rules. Every triple then has a three-literal body. Two or more triples, one nonalternating, already save four at its fixed start parity. With two alternating triples the remaining budget is (1,1). One alternating triple leaves (2,2). One nonalternating triple leaves zero at its start parity and two at the other. Zero triples leave (3,3).

Available length-two and length-three words	Budget (even, odd)	Farthest boundary
None	(3, 3)	10
010	(2, 2)	6
101	(2, 2)	7
010, 101	(1, 1)	5
01, 010, 101	(1, 1)	9
10, 010, 101	(1, 1)	13
00	(1, 1)	2
11	(1, 1)	4
001	(2, 0)	1
011	(0, 2)	4
100	(0, 2)	0
110	(2, 0)	5

The certificate on the next page computes the last column using only the first 18 bits. A complete grammar would have to reach a boundary at or beyond 14. Even the farthest case stops at 13.

The 18-bit prefix certificate

All T_n for n ≥ 5 start with the following prefix. Work with tile starts i < 14. All tiles of length at most four and all five-bit windows around these starts fit within the prefix whenever their left boundary is nonnegative.

```
011010011001011010
```

Choose the length-four expansions

For each distinct four-bit factor in the prefix, its phase is fixed. Try each of its eight partitions at the three internal cuts. Each block must be a literal or one of the row's short words. Charge each literal at its phase. For each factor either omit it or include it at the cost of one valid partition, subtracting the charge from the budgets. A zero-cost factor can always be allowed.

Describe a literal from a longer rule

One literal slot in a rule of expansion length at least five lies in some five-bit window within that expansion. The window and the slot's offset are fixed each time this rule is used. For each parity, enumerate all masks of positions i < 14 defined by a window and offset. A slot is allowed at i if that window appears at the specified offset around i. Take unions of at most as many masks as the remaining literal budget for that parity.

Check which prefix boundaries can be reached

Unfold the grammar down to expansions of lengths two through four and literals. Its resulting tiling is one of the possibilities we allow. Length-four tiles use the selected vocabulary. Every remaining literal came from a rule of expansion length at least five, so its position belongs to the union of the corresponding slot masks. Extra allowed tiles or positions can only enlarge reachability.

```
reached = {0}
for i = 0, ..., 13, in order:
    if i is reached and allowed as a literal, mark i + 1
    if i is reached, mark i + len(w) for each matching tile w
```

Take the largest marked boundary over all vocabulary and mask choices in each row. The integer replay checks 8,829 combinations after merging identical permission masks. Every row stops below 14. This contradicts the tiling of a full word and proves that some D_p is at least four.

The base cases and the encoder

Exact minima at small levels

In a minimum grammar each non-start rule is referenced at least twice. Its expansion therefore occurs twice disjointly in the word. Choose all subsets of the distinct candidate words of length at least two with this property. For each dictionary, find the shortest body of each chosen rule and the start rule by segmenting its word into literals and strictly shorter dictionary words. These choices are independent and acyclic. The minimum total over dictionaries is the grammar minimum.

Level	Candidate words	Dictionaries checked	Minimum
0	0	1	1
1	0	1	2
2	0	1	4
3	2	4	7
4	12	4,096	11

For $n \geq 5$ the reduction gives $g_n \geq g_{n-1} + 4$. Starting at $g_4 = 11$ gives $g_n \geq 4n - 5$ for $n \geq 4$. The exact base check gives the same formula at $n = 3$.

A grammar that attains the bound

For T_3, use $C \to 01$ and $S \to C1C0C$, of total cost seven. To apply another substitution, replace literal 0 and 1 in all old rule bodies by references to new rules for 01 and 10, respectively. Old body lengths stay fixed and the two new rules cost four. Iterating gives the claimed upper bound.

Reproduction

The source bundle includes a grammar encoder and decoder, the finite checker and a Lean 4 cross-check of the obstruction table, exact base cases and numerical induction. The proof of the structural reduction is given on pages 2-4. Both encoder variants were checked against the independent bit-count definition at levels 0 through 20.

The binary grammar minimum

The paper that raised the question counts productions in Chomsky normal form [1]. Each reachable rule emits one terminal or joins two earlier variables. A word with both bits needs two terminal rules, which can be shared across all uses.

Long factors occur only once

For n ≥ 2 no factor longer than a quarter of T_n occurs twice. At levels 2 and 3, the length-two and length-three windows, respectively, are all distinct. For n ≥ 4, take two occurrences of a factor of length $2^{n-2} + 1$. This length is at least five, so both starts have the same parity. Keep that parity. The occurrences project to two distinct starts of the same factor of length $2^{n-3} + 1$ in T_{n-1} or its complement, contradicting induction. Any longer repeat would have this length as a repeated prefix.

Two single-use rules are unavoidable

In the unfolded binary grammar, stop expanding each branch at the first node with length at most a quarter of the word. These leaves cover the word, so there are at least four. The truncated full binary tree has at least three internal nodes, including the root. Their expansions exceed a quarter of the word. Each therefore occurs once and has a distinct rule. At least two such non-start rules have only one reference in the grammar.

Let r be the number of binary concatenation rules. Substitute the terminal productions into their references to give an unrestricted grammar of cost 2r. Inline two of the single-use rules, saving one symbol each. For n ≥ 3 the symbol minimum then gives

$$4n - 5 \le 2r - 2, \qquad r \ge 2n - 1.$$

At level 2 the same argument uses $g_2 = 4$ and gives r ≥ 3. At level 1, joining the two terminal rules needs one binary rule. The matching construction stores T_j and its complement for j = 1 through n−1, each from one binary rule, then adds one start rule. Including both terminal productions gives 2n+1. At level zero the start is the sole terminal rule.

[1] T. Mieno, S. Inenaga and T. Horiyama. *RePair Grammars Are the Smallest Grammars for Fibonacci Words*. CPM 2022, §6. 10.4230/LIPIcs.CPM.2022.26.